The Rental Penalty: Are Market-Rate Tenants More Exposed to Housing Cost Overburden in Europe?

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1 Project information

Project title: The Rental Penalty: Are Market-Rate Tenants More Exposed to Housing Cost Overburden in Europe?

Group members: Add names and student numbers here

Tutorial group: Add tutorial group number here

Tutorial lecturer: Add tutorial lecturer name here

GitHub repository: Add public GitHub repository link here

2 Part 1 - Identify a Social Problem

2.1 1.1 Describe the social problem

Housing affordability is a major social issue because housing is a basic necessity and one of the largest fixed costs for households. However, housing cost pressure is not experienced equally by all households. People who own their home and people who rent their home are in different economic positions.

Homeowners may build wealth through ownership, and some homeowners may see their housing costs decline over time. Renters, especially tenants paying market rent, usually do not build housing wealth and may be more directly exposed to high rents and rental market pressures.

This project therefore focuses on a concrete housing inequality question: whether **market-rate tenants face a higher housing cost burden than homeowners**.

Main research question:

Do market-rate tenants face a higher housing cost burden than homeowners across European countries between 2014 and 2024?

Second-step analysis:

After measuring this renter-homeowner difference, we examine whether countries with a larger rental penalty also have higher poverty or social exclusion risk.

The goal is not to prove that renting causes poverty. Instead, we measure whether housing cost pressure is distributed unequally between tenure groups, and whether this inequality is connected to broader poverty and social exclusion patterns.

2.2 1.2 Relevance of the problem

This topic is relevant because a high housing cost burden can reduce the amount of income households have left for other essential needs, such as food, transport, healthcare, education and energy. If this burden is much higher for renters than for homeowners, housing affordability is not just a general economic issue, but also a distributional issue within the housing market.

The project contributes by measuring the **rental penalty**: the difference between the housing cost overburden rate of market-rate tenants and homeowners with a mortgage or loan.

In this project, the rental penalty is defined as:

$$Rental\ Penalty = Overburden\ rate\ of\ market\text{-}rate\ tenants - Overburden\ rate\ of\ owners\ with\ mortgage/loan$$

A positive rental penalty means that market-rate tenants are more likely to be overburdened by housing costs than homeowners with a mortgage or loan.

We compare market-rate tenants with owners who still have a mortgage or loan because both groups usually face active monthly housing payments. Comparing tenants with owners without a mortgage or loan would likely exaggerate the gap, because many owners without a mortgage have much lower ongoing housing costs.

3 Part 2 - Data Sourcing

3.1 2.1 Data sources

We use public data from Eurostat. Eurostat is the statistical office of the European Union and provides comparable data across European countries.

We use two main datasets:

1. **Housing cost overburden rate by tenure status**
   Eurostat code: `ilc_lvho07c`
2. **Persons at risk of poverty or social exclusion by age and sex**
   Eurostat code: `ilc_peps01n`

The first dataset is used to compare housing cost overburden between tenure groups. The second dataset is used in the second-step analysis to examine whether countries with a larger rental penalty also have higher poverty or social exclusion risk.

```
## indexed 0B in  0s, 0B/sindexed 1.00TB in  0s, 4.54PB/s
```

```
## indexed 0B in  0s, 0B/sindexed 1.00TB in  0s, 195.69TB/s
```

3.2 2.2 Short summary of the datasets

The housing dataset contains country-year observations by tenure status. The poverty/social exclusion dataset contains country-year observations by age and sex.

dataset	eurostat_code	rows	columns
Housing cost overburden rate by tenure status	ilc_lvho07c	3925	6
Persons at risk of poverty or social exclusion by age and sex	ilc_peps01n	87720	7

3.3 2.3 Description of variables

The most important variables in the housing dataset are:

- `geo`: country code;
- `TIME_PERIOD`: year;
- `tenure`: housing tenure group;
- `values`: housing cost overburden rate as a percentage.

The most important variables in the poverty dataset are:

- `geo`: country code;
- `TIME_PERIOD`: year;
- `age`: age group;
- `sex`: sex;
- `values`: at-risk-of-poverty or social exclusion rate as a percentage.

The data are socio-economic indicators and are suitable for comparing countries over time. A limitation is that these are country-level indicators, meaning that we can compare broad national patterns but not individual household situations.

4 Part 3 - Quantifying

4.1 3.1 Data cleaning

We focus on the period 2014-2024. For the housing dataset, we use the following tenure groups:

- `OWN_L`: owners with mortgage or loan;
- `OWN_NL`: owners without mortgage or loan;
- `RENT_MKT`: tenants renting at market price;
- `RENT_FR`: tenants renting at reduced price or free;
- `TOTAL`: total population.

For the poverty/social exclusion dataset, we use the total population and both sexes combined.

4.2 3.2 Generate necessary variables

The main new variable is the **rental penalty**. To calculate it, we reshape the housing dataset so that each tenure group becomes its own column.

observations	countries	first_year	last_year	average_rental_penalty
345	36	2015	2024	22.35

A positive `rental_penalty` means that market-rate tenants face a higher housing cost overburden rate than homeowners with a mortgage or loan.

5 Part 4 - Results

5.1 4.1 Sub-population variation: housing burden by tenure group

First, we compare tenure groups directly. This shows whether market-rate tenants are more exposed to housing cost overburden than homeowners.

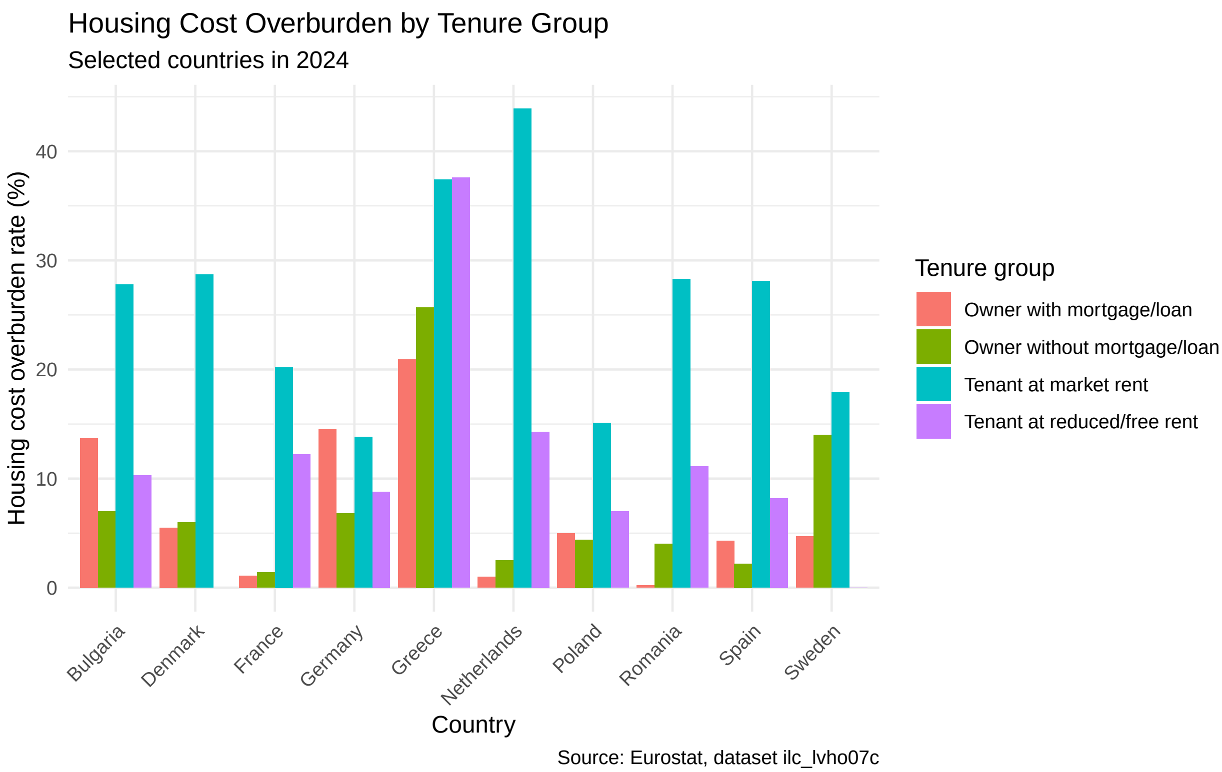
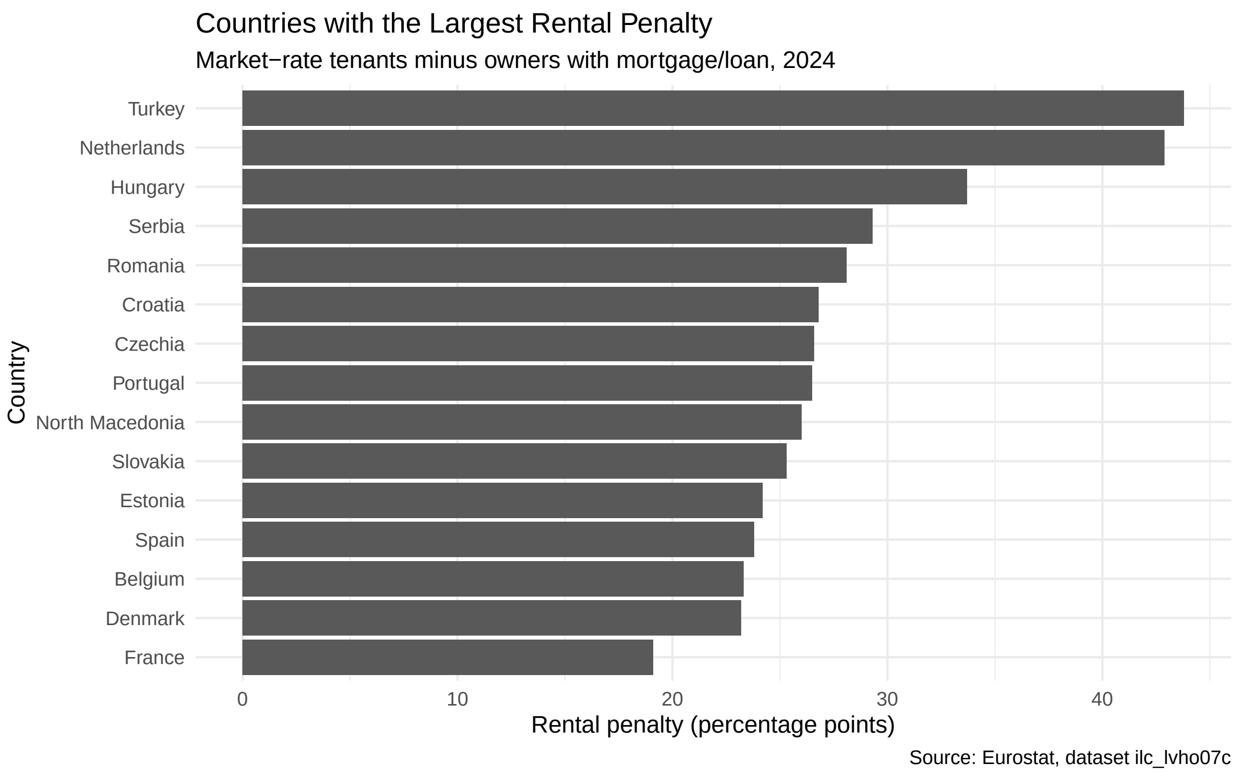


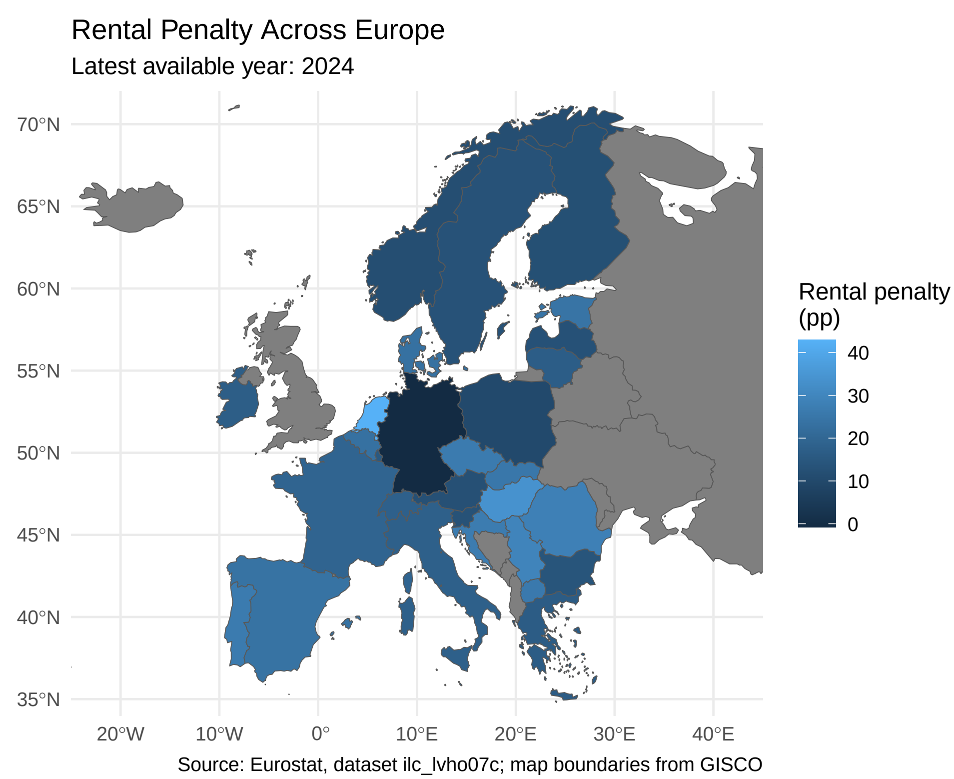
Figure 1 shows that across almost all selected countries, market-rate tenants consistently face the highest housing cost overburden rate. This pattern holds regardless of whether the country is economically more developed (Netherlands, Sweden, France) or less developed (Bulgaria, Romania, Poland), suggesting that tenure-based housing inequality is a broad European phenomenon rather than one limited to specific national contexts. Greece remains the notable exception, where overburden rates are high across all tenure groups, indicating a systemic affordability crisis beyond the rental market alone.

5.2 4.2 Spatial variation: where is the rental penalty largest?

We now identify the countries with the largest rental penalty in the latest available year.



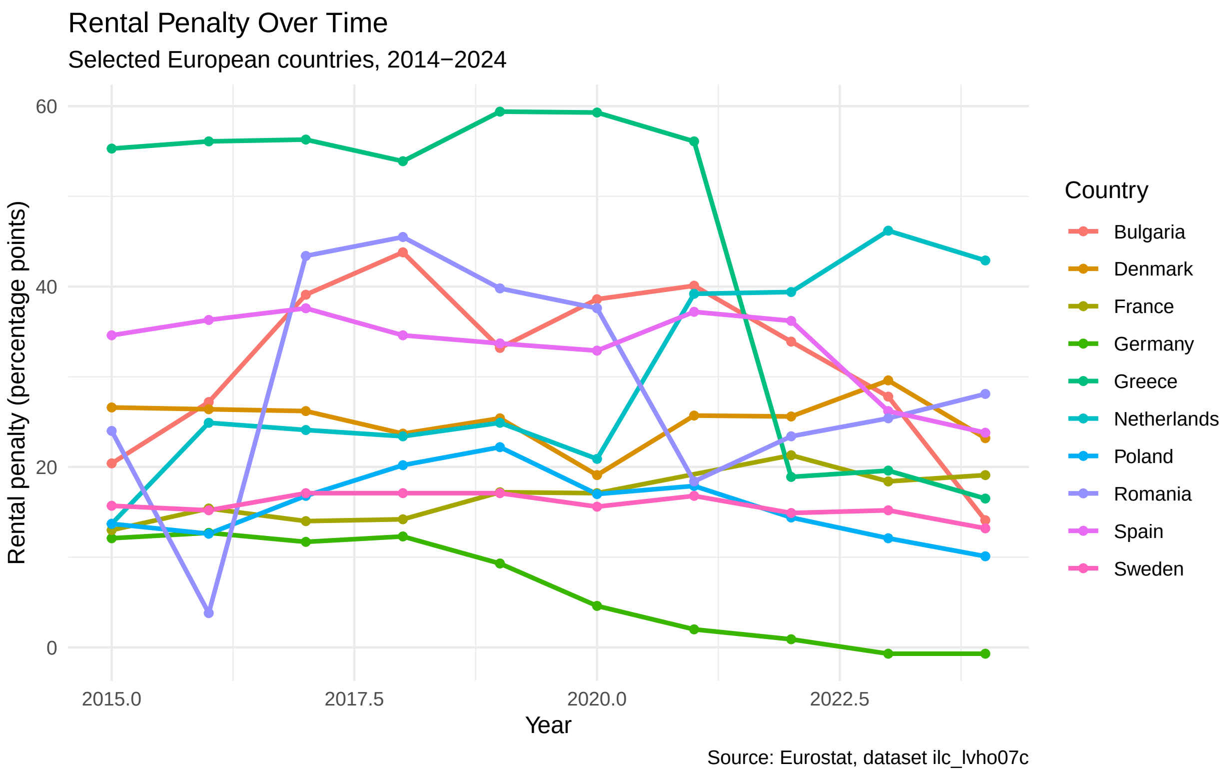
A high rental penalty does not necessarily mean that all housing costs are highest in that country. It means that the difference between market-rate tenants and homeowners with a mortgage or loan is largest.



The map visualises the spatial distribution of the rental penalty across Europe. It shows that the renter-homeowner burden gap is not evenly distributed, but differs substantially between countries.

5.3 4.3 Temporal variation: did the rental penalty change over time?

Next, we examine whether the rental penalty has changed over time in selected countries.

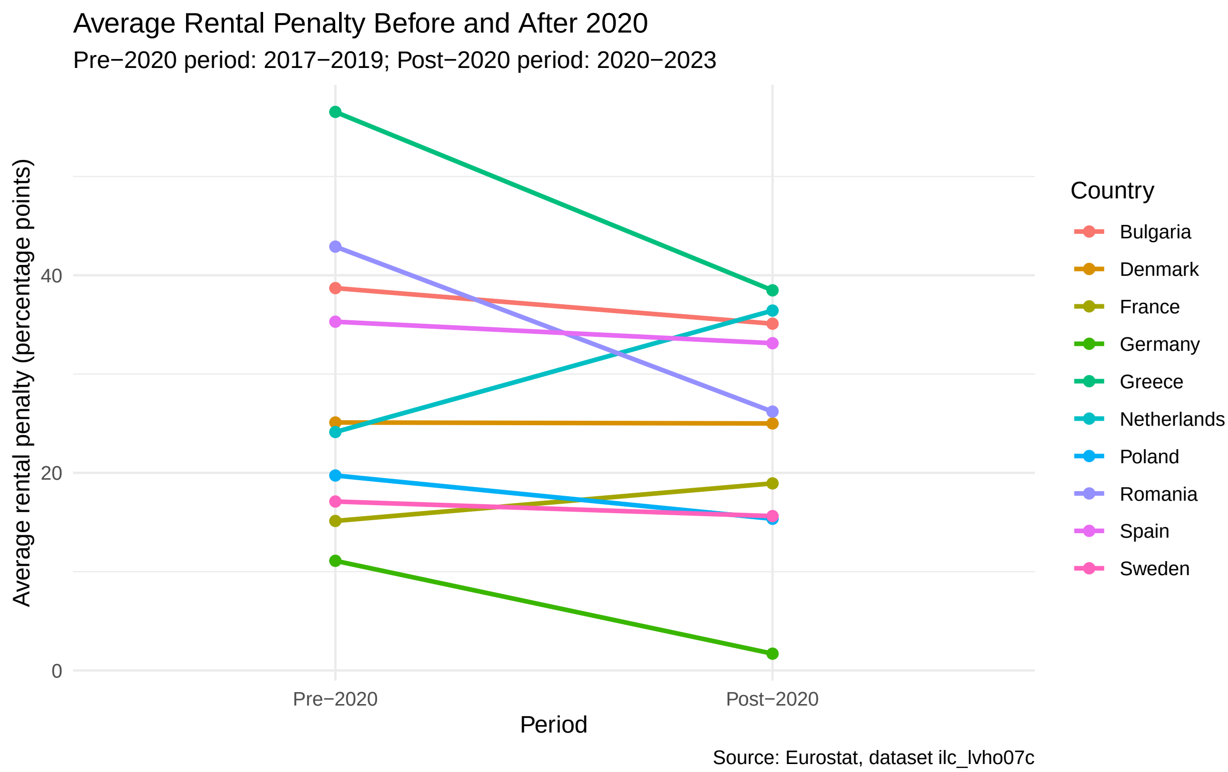


The rental penalty is not stable across countries. Some countries show a declining gap, while others show a persistent or increasing gap. This suggests that the rental penalty is not only a general European trend, but may depend on national housing market structures, rental market regulation and the relative position of homeowners and renters. Western European countries such as the Netherlands and France show a different trend pattern compared with Eastern European countries such as Romania and Bulgaria, suggesting that national housing market development may shape how the rental penalty evolves over time.

5.4 4.4 Event association: did the rental penalty change after 2020?

We use 2020 as an event point. This is not a causal test of COVID-19. It helps us examine whether the rental penalty changed around the period of COVID-19 and the later cost-of-living pressure.

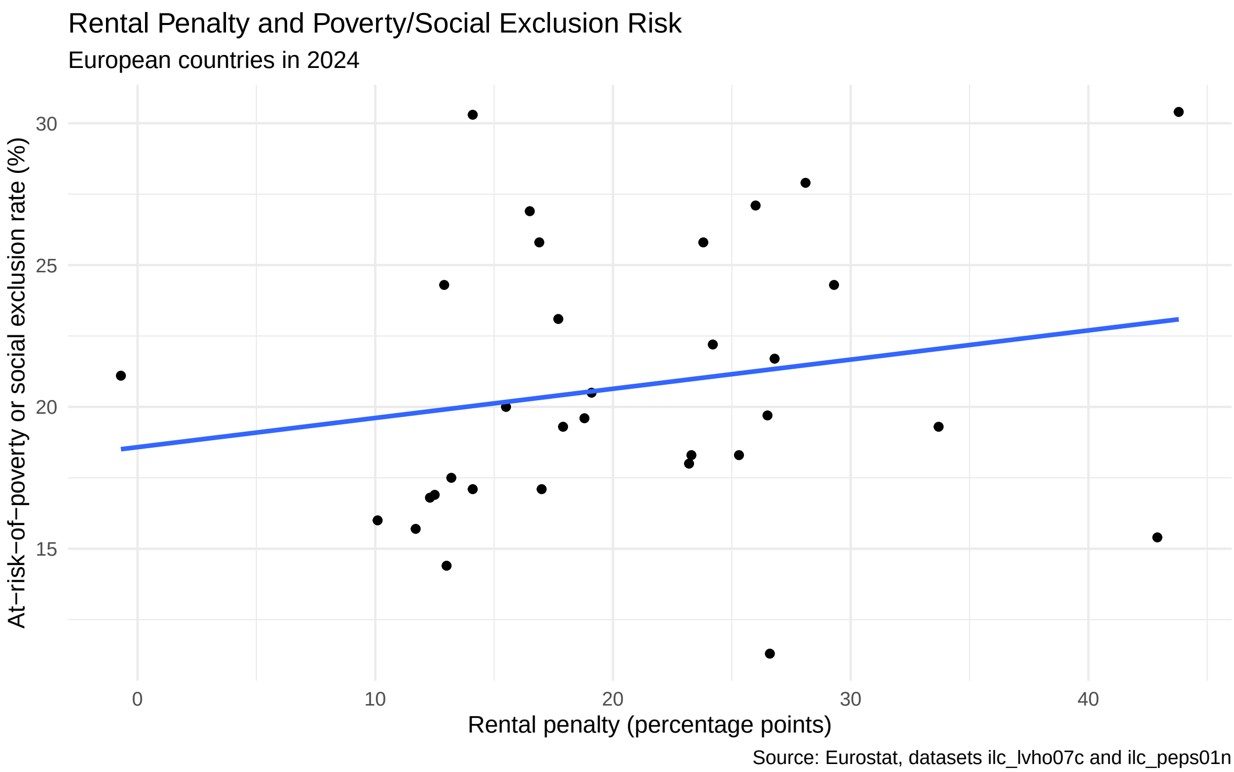
Because housing markets respond slowly and policy measures differed across countries, we should not expect one uniform post-2020 effect.



A noticeable change from pre-2020 to post-2020 is the decline in the average rental penalty in most of the countries selected. Both economically developed and less developed european countries show a decrease in the metric, with the Netherlands being an exemption, showing a strong increase in the rental penalty. This plot shows whether the rental penalty changed around 2020. We do not claim that 2020 caused the change. The graph should be interpreted as descriptive event evidence.

5.5 4.5 Second-step analysis: rental penalty and poverty risk

Finally, we examine whether countries with a larger rental penalty also have higher poverty or social exclusion risk.



The relationship between rental penalty and poverty/social exclusion risk should be interpreted carefully. If the trend line is positive, this suggests that countries with a larger renter-homeowner burden gap may also face higher poverty risk. However, the spread of points matters: if countries are widely scattered, then rental market inequality and poverty risk do not fully overlap.

6 Part 5 - Discussion

6.1 5.1 Discuss your findings

The analysis suggests that market-rate tenants often face a higher housing cost burden than homeowners with a mortgage or loan. This supports the main idea of the project: housing affordability pressure is not distributed equally across tenure groups.

The size of the rental penalty differs strongly across Europe. Some countries show a much larger gap between market-rate tenants and homeowners than others. This means that housing affordability problems cannot be understood only by looking at the general level of housing costs. It is also important to examine who carries the burden.

The over-time analysis shows that the rental penalty does not follow the same path in every country. In some countries the gap declines, while in others it remains high or increases. This suggests that the rental penalty may be shaped by national housing systems, rental market regulation, income dynamics and the relative position of renters and homeowners.

The second-step analysis links the rental penalty to poverty/social exclusion risk. If the relationship is positive, this suggests that housing tenure inequality may be part of broader social inequality. If the relationship is weak, this still provides an important insight: rental market structures and housing policy may matter independently from general poverty levels.

The main limitation is that this project is descriptive. It does not prove that renting causes poverty or that poverty causes renting. The analysis shows patterns and associations at country level. Future research could use household-level data, include housing policy indicators, or compare cities and regions rather than only countries.

7 Part 6 - Reproducibility

7.1 6.1 GitHub repository link

Provide the link to your public repository here:

[add GitHub repository link]

7.2 6.2 Reference list

Eurostat. (2026). *Housing cost overburden rate by tenure status* [Data set]. Eurostat. Dataset code: `ilc_lvho07c`.

Eurostat. (2026). *Persons at risk of poverty or social exclusion by age and sex* [Data set]. Eurostat. Dataset code: `ilc_peps01n`.

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